

Bridging the Generalist–Subspecialist Gap with GPT-5-Thinking: Dual-Center Evaluation in Orbital and Head-and-Neck Tumor MRI Reports

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Background: Identifying tumor histologic subtypes in complex regions is challenging for generalist radiologists. Whether reasoning large language models can bridge this expertise gap by interpreting radiologic descriptions remains underexplored.

Purpose: To evaluate whether Generative Pretrained Transformer (GPT)-5-Thinking can perform similarly to subspecialists and help bridge the generalist expertise gap in interpretation of MRI scans of orbital and head-and-neck tumors.

Materials and Methods: This dual-center retrospective study was performed at a generalist practice center and a subspecialist practice center and included convenience series of patients with pathologically confirmed orbital or head-and-neck tumors who had pretreatment MRI reports. GPT-5-Thinking processed narrative MRI reports to output benign–malignant classification and top-3 differential diagnoses. Phase 1 compared model accuracy with routine clinical report diagnoses in generalist and subspecialist practice settings. In Phase 2, seven generalists interpreted selected tumor reports twice, without and with GPT-5-Thinking assistance. McNemar tests and mixed-effects logistic regression models were used for analysis.

Results: This study included 1000 patients (mean age $\pm$ SD, 51 years $\pm$ 17.0; 520 men). In phase 1, the top-1 accuracy of GPT-5-Thinking was similar to that of subspecialists in patients with orbital tumors (76.7% [230 of 300] vs 78.3% [235 of 300]; $P = .64$) and head-and-neck tumors (67.5% [135 of 200] vs 68.0% [136 of 200]; $P = .92$). GPT-5-Thinking had higher top-1 accuracy than did routine generalists for patients with orbital tumors (74.0% [222 of 300] vs 68.7% [206 of 300]; $P = .03$) and head-and-neck tumors (64.0% [128 of 200] vs 27.0% [54 of 200]; $P < .001$). In phase 2, GPT-5-Thinking assistance was associated with an increase in generalist top-1 diagnostic accuracy from 61.4% (430 of 700) to 70.3% (492 of 700) ($P < .001$) in patients with orbital tumors and from 47.0% (329 of 700) to 61.0% (427 of 700) ($P < .001$) in patients with head-and-neck tumors.

Conclusion: Using MRI reports of patients with orbital or head-and-neck tumors, Generative Pretrained Transformer (GPT)-5-Thinking showed diagnostic accuracy similar to that of subspecialists; GPT-5-Thinking assistance was also associated with higher generalist diagnostic accuracy.

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Diagnostic imaging frequently confronts a long-tail distribution in which a few common entities dominate routine volume, whereas numerous low-prevalence, high-complexity conditions are encountered only sporadically (1,2). For generalist radiologists, limited exposure to these rare phenotypes often leads to broader, less specific differential diagnoses and substantial diagnostic variability (3,4). In contrast, subspecialists accrue concentrated experience that supports more confident, actionable impressions and standardized care (5,6). This persistent expertise gap represents a critical clinical vulnerability; diagnostic ambiguity in specialized domains can delay definitive management and contribute to disparate patient outcomes.

Artificial intelligence has been proposed to support generalist readers, but most deployed tools remain image-centric algorithms limited to narrow, task-specific applications (7). Although these models excel at binary classification, they often struggle to emulate the multivariate reasoning required for subtype-level differential diagnoses in complex settings (8,9). Moreover, many image-based systems function as black boxes, offering limited interpretability and requiring bespoke integration that can hinder

routine clinical adoption (10,11). There is an urgent need for an interpretable assistant that can bridge the expertise gap by synthesizing complex findings into auditable clinical logic.

Large language models offer a workflow-concordant solution by emulating the high-level semantic reasoning of a subspecialist. By processing routine radiology report descriptions, they can ingest the same semantic cues used by experts—such as growth patterns, enhancement kinetics, and clinical tempo—to generate a ranked differential with explicit supporting rationales (12,13). Unlike pixel-based models, these language models operate directly within the text-based reporting environment, facilitating transparency and audibility (14,15). However, large-scale evidence is lacking regarding their ability to handle high-complexity tasks within native reporting workflows.

This study aims to evaluate whether Generative Pretrained Transformer (GPT)-5-Thinking can perform similarly to subspecialists and help bridge the generalist expertise gap in interpretation of MRI scans of orbital and head-and-neck tumors. First, a deep-dive evaluation in an orbital tumor group was performed to establish diagnostic accuracy against subspecialist standards; subsequently, a

Abbreviation

GPT = Generative Pretrained Transformer

Summary

With use of MRI reports of orbital or head-and-neck tumors, Generative Pretrained Transformer-5-Thinking showed diagnostic accuracy similar to that of subspecialists, and its use was associated with higher generalist diagnostic accuracy.

Key Results

- In this retrospective dual-center study of 1000 patients with MRI-assessed orbital or head-and-neck tumors, the top-1 accuracy of Generative Pretrained Transformer (GPT)-5-Thinking was similar to that of subspecialists (76.7% [230 of 300] vs 78.3% [235 of 300]; $P = .64$) for orbital tumors, but the top-1 accuracy was higher than that of routine generalists (74.0% [222 of 300] vs 68.7% [206 of 300]; $P = .03$).
- GPT-5-Thinking assistance was associated with an increase in generalist accuracy from 61.4% (430 of 700) to 70.3% (492 of 700) ($P < .001$) and an increase in interreader agreement (Fleiss $\kappa = 0.41$ – 0.71 ; $P < .001$) in patients with orbital tumors.
- In patients with head-and-neck tumors, GPT-5-Thinking assistance was associated with an increase in generalist accuracy from 47.0% (329 of 700) to 61.0% (427 of 700) ($P < .001$), and GPT-5-Thinking showed accuracy similar to that of subspecialists (67.5% [135 of 200] vs 68.0% [136 of 200]; $P = .92$).

large-scale extension to a heterogeneous head-and-neck group was conducted to assess whether the framework retained similar accuracy patterns across an expanded spectrum of pathologic conditions.

Materials and Methods

Study Design

This retrospective diagnostic accuracy study was conducted at two tertiary teaching hospitals with distinct clinical profiles. The Second Hospital of Jilin University (center 1) represents a generalist setting, where orbital and head-and-neck MRI interpretation is secondary to a routine, multisystem workload. In contrast, Beijing Tongren Hospital at Capital Medical University (center 2) is a specialized setting, where such imaging is central to its radiologists' practice. Both institutional review boards approved the study and waived informed consent because of its retrospective design. Reporting followed the Standards for Reporting of Diagnostic Accuracy Studies–Artificial Intelligence (16).

The evaluation of GPT-5-Thinking based on MRI reports was structured into two phases. In the first phase (benchmarking), it was compared against the original routine clinical diagnoses at each center. In the second phase (a reader study), it was evaluated as an aid to generalist radiologists. This two-phase framework was applied to two distinct patient groups. The primary analysis focused on the orbital tumor group. The head-and-neck tumor group extended the evaluation to broader anatomic regions: the larynx or pharynx, major salivary glands, sinonasal region or nasopharynx, and deep neck spaces or skull base. This group was assessed using the same framework, adapted for these regions. A study design schematic is provided in Figure S1.

Both groups were assembled independently as convenience series, spanning an overall study period from January 2018 to August 2025. For each group at each center, full-text searches of reports in the radiology information system and picture archiving and communication system were conducted. Reports

were screened in reverse-chronologic order until the prespecified per-center target was reached.

Eligibility criteria were consistent across both groups. Patients were included if an MRI report documented a mass in the respective anatomic region (orbital or head and neck). Patients were excluded if (a) histopathologic confirmation was unavailable, nondiagnostic, indeterminate, or discordant with the index mass or (b) only posttreatment or follow-up MRI reports were available, or the available pretreatment MRI report was not the earliest one for that patient. Figure 1 details the patient selection process and the specific study periods for each center and group.

MRI Scan Acquisition

MRI examinations were performed using 3.0-T systems, including Discovery MR750 (GE HealthCare), Ingenia (Philips Healthcare), MAGNETOM Skyra (Siemens Healthineers), and uMR 790 (United Imaging Healthcare). When contrast-enhanced MRI was performed, gadopentetate dimeglumine (Magnevist; Bayer Healthcare) was administered intravenously at a dose of 0.1 mmol/kg body weight. Acquisition protocols varied by center and anatomic region and are detailed in Appendix S1.

Image Analysis

The GPT-5-Thinking application programming interface was used (Open AI, v5.2; December 11, 2025) with deterministic decoding (temperature = 0). The model-selection and imaging-modality rationales are provided in Appendix S2. The model was evaluated in a frozen, stateless, and single-turn setting—precluding any local optimization or cross-report learning effects. Reference-standard pathologic results were withheld throughout inference.

A standardized system prompt was iteratively refined by two senior head-and-neck radiologists (J.X. and X.Q., both with >15 years of experience) to emulate a subspecialist-style consultation. For each report, the model was instructed to (a) provide a benign–malignant classification and (b) output ranked top-3 differential diagnoses with concise, report-grounded reasoning for each candidate. The final prompt template was established on a separate prompt-development set, held out from the evaluation data and locked before evaluation (Fig 2).

Inputs were strictly synchronized with information available at the initial interpretation. Each input integrated clinical context from the original requisition and the MRI report *Findings* section. To prevent label leakage and retrospective bias, all diagnostic impressions were manually removed and any postimaging clinical notes or histopathologic results were excluded. CT-specific details were omitted for patients with concurrent studies. A 3rd-year radiology resident-in-training (Y.Z.), blinded to the reference standard, entered all data into a fixed template.

Outcomes

Histopathologic findings obtained from surgical resection or core biopsy served as the reference standard for all diagnostic outcomes. Primary outcomes were accuracy of benign–malignant classification and top-1 accuracy. Secondary outcomes included top-2 and top-3 accuracy, report reading time, interreader agreement (Fleiss κ), and report quality scores. Report quality was assessed using a prespecified, five-dimension, five-point Likert scale. Full details of measurement definitions and assessment procedures are provided in Appendix S3.

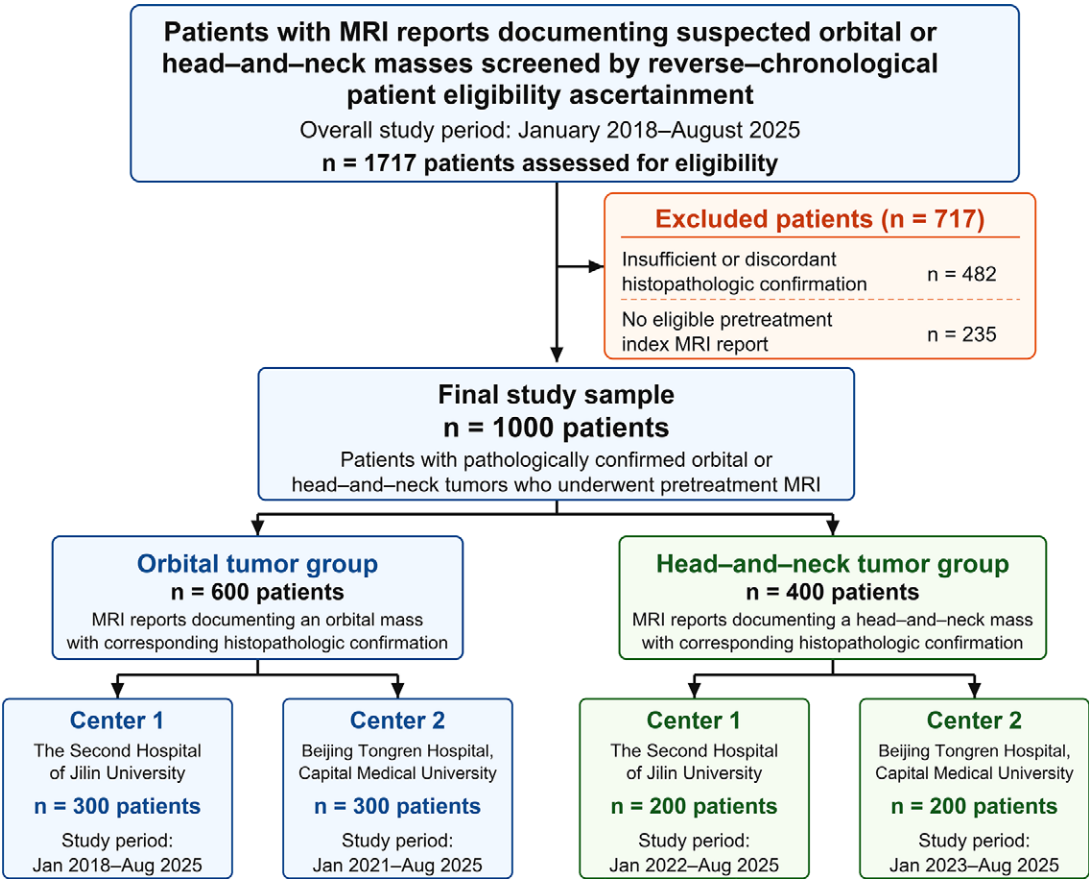


Figure 1: Patient selection flow diagram. Flow diagram of reverse-chronologic patient eligibility ascertainment for the orbital tumor group and head-and-neck tumor group at two centers, from January 2018 to August 2025. The orbital tumor group comprised patients with MRI reports documenting an orbital mass with corresponding histopathologic confirmation. The head-and-neck tumor group comprised patients with MRI reports documenting a head-and-neck mass with corresponding histopathologic confirmation. Patient eligibility ascertainment proceeded in reverse-chronologic order until the prespecified per-center study sample size was reached. The final study sample included 1000 patients: 600 in the orbital tumor group and 400 in the head-and-neck tumor group.

Phase 1: Benchmarking against routine clinical practice

For each center, the benign–malignant classification and the top-1 accuracy were extracted from the routine clinical report to represent real-world reader accuracy (generalists at center 1; subspecialists at center 2). GPT-5-Thinking independently processed the corresponding textual inputs for the same reports. Because routine clinical practice reports usually record a single working diagnosis, human top-2 and top-3 differentials could not be reliably reconstructed and were not compared.

Phase 2: Paired reader-aid study

To compare reader accuracy with and without GPT-5-Thinking while controlling for tumor-type distribution, a prespecified stratified sample of 100 reports was assembled from the retrospective pool. Seven generalists (L.D., T.G., Y.Y., Y.L., Q.H., G.C., and H.L.; 7–15 years of general diagnostic radiology experience) from center 1 interpreted all 100 reports twice: unaided and after a 4-week or longer washout with randomized report order per session, aided by GPT-5-Thinking. Per report, readers recorded a benign–malignant classification, provided a top-3 differential with a single top-1 selection, and captured reading time using an on-screen timer. During the aided session, readers could access and use the model’s outputs and rationales in any sequence or manner they preferred; final responsibility for the

report remained with the reader. Detailed reader study settings are provided in Tables S8 and S9.

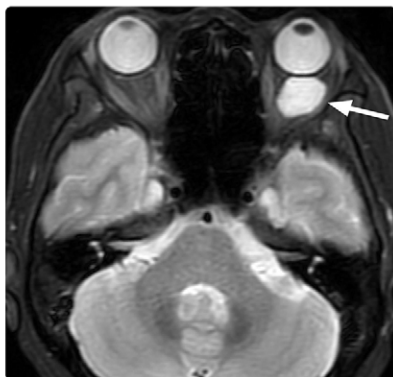
Extended Diagnostic Evaluation

To rigorously test the inferential limits and robustness of the framework within the orbital group, a four-tiered extended evaluation was conducted: (a) a category analysis was performed to evaluate discrimination accuracy across the full spectrum of orbital abnormalities via benign–malignant confusion matrices and class-wise heatmaps; (b) two sensitivity analyses were performed, one restricting to the overlapping accrual window (January 2021–August 2025) to confirm within-center stability and one excluding the two most common entities to test resilience across the long tail; (c) a performance ceiling analysis was executed via subspecialist error re-evaluation combined with post hoc audits to isolate the effect of input granularity; and (d) a qualitative analysis of representative MRI reports—spanning both successful corrections and residual errors—was performed to provide insight into the model’s reasoning behaviors and typical failure modes.

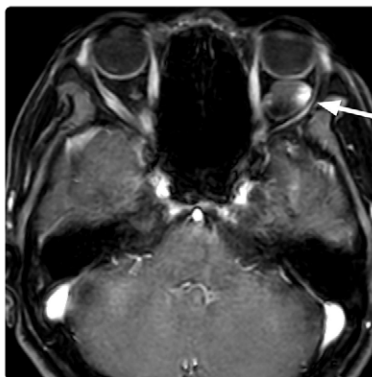
Statistical Analysis

Continuous variables were summarized as mean ± SD for normally distributed data and as median (IQR, range) for skewed data; categorical variables were summarized as counts and

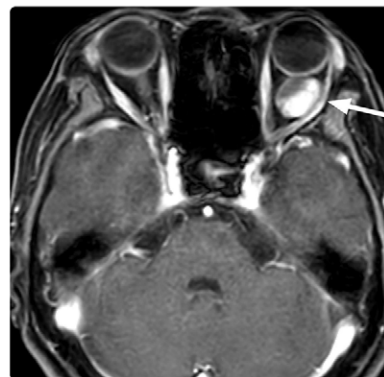
A Representative MRI



A1. Axial T2-weighted fat-suppressed
Well-circumscribed hyperintense lesion in the left intraconal space posterior to the globe.



A2. Axial T1-weighted early post-contrast
Early peripheral nodular enhancement of the lesion.



A3. Axial T1-weighted delayed post-contrast
Progressive centripetal fill-in with near complete enhancement.

B Fixed prompt

You are an orbital imaging radiologist. Based on the patient's clinical information and the MRI findings provided, determine whether the intraorbital lesion is benign or malignant. Provide a ranked list of the three most likely diagnoses (with specific disease names) and briefly summarize the supporting evidence for each from the imaging features and clinical data.

C Inputs

Clinical History

Female, 55 years old; left orbital mass detected during physical examination for 20 days.

MRI Findings

Quasi-round $20 \times 14 \times 16$ mm lesion in the left intraconal region posterior to the globe. Isointense on T1-weighted imaging, hyperintense on T2-weighted imaging, and high signal on DWI. Marked enhancement with early peripheral nodular enhancement and progressive centripetal fill-in. Optic nerve compressed and displaced nasally. Lateral rectus muscle mildly displaced outward. No abnormal signal in adjacent orbital bones.

D Output

Benign vs malignant assessment: favors benign.

Rationale: The lesion is intraconal, round, well-circumscribed, and demonstrates a characteristic dynamic enhancement pattern of early peripheral nodular enhancement with progressive centripetal fill-in. It displaces rather than infiltrates adjacent structures (optic nerve and lateral rectus), shows high DWI signal without features of aggressive malignancy, and there is no adjacent bone abnormality. These imaging characteristics are most consistent with a benign vascular lesion.

Top three most likely diagnoses

1. Cavernous venous malformation (cavernous hemangioma)

Support: Typical intraconal location in a middle-aged woman; well-circumscribed; T2 hyperintense; early peripheral nodular enhancement with centripetal fill-in; causes displacement rather than invasion; normal adjacent bone.

2. Nerve sheath tumor

Support: Can be intraconal with avid enhancement and may displace the optic nerve; however, enhancement is usually more homogeneous or shows a target sign, and lacks the classic centripetal fill-in pattern.

3. Orbital lymphoma

Support: DWI hyperintensity can be seen in lymphoma and patient age is compatible; however, lymphoma more often shows diffuse or plaque-like involvement with extraconal predominance and typically does not demonstrate centripetal fill-in.

Conclusion

The imaging and clinical findings most strongly support a benign intraconal vascular lesion, with cavernous venous malformation (cavernous hemangioma) as the leading diagnosis; nerve sheath tumor is a secondary consideration, and orbital lymphoma is less likely.

Figure 2: Representative report-to-output workflow for Generative Pretrained Transformer (GPT)-5-Thinking interpretation of orbital MRI scan. This figure shows how deidentified clinical context and MRI report findings from one pretreatment patient-level report input were converted into a structured GPT-5-Thinking diagnostic output. **(A)** Images in a 55-year-old woman with a left orbital mass detected during physical examination. Axial fat-suppressed T2-weighted MRI scan shows a well-circumscribed hyperintense lesion in the left intraconal space posterior to the globe (arrow). Axial early and delayed postcontrast T1-weighted MRI scans show early peripheral nodular enhancement with progressive centripetal fill-in. **(B)** Fixed prompt used for GPT-5-Thinking interpretation. **(C)** Deidentified clinical history and MRI report findings used as model inputs. **(D)** Structured model output, including benign–malignant classification, ranked differential diagnosis, rationale, and conclusion.

Table 1: Patient Characteristics of the Final 1000-Patient Study Sample

Characteristic	All Patients (<i>n</i> = 1000)	Orbital Tumor Group (<i>n</i> = 600)	Head-and-Neck Tumor Group (<i>n</i> = 400)
Age (y)	51 ± 17.0 (18–90)	45 ± 12.8 (18–79)	60 ± 18.7 (18–90)
Sex			
Male	520 (52.0)	330 (55.0)	190 (47.5)
Female	480 (48.0)	270 (45.0)	210 (52.5)
Center			
Center 1	500 (50.0)	300 (50.0)	200 (50.0)
Center 2	500 (50.0)	300 (50.0)	200 (50.0)
Tumor classification			
Benign	634 (63.4)	459 (76.5)	175 (43.8)
Malignant	366 (36.6)	141 (23.5)	225 (56.2)

Note.—Unless otherwise indicated, data are numbers of patients, with percentages in parentheses. Age data are means ± SDs, with ranges in parentheses. Center 1 refers to the Second Hospital of Jilin University; center 2 refers to Beijing Tongren Hospital, Capital Medical University.

Table 2: Phase 1 Diagnostic Metrics for GPT-5-Thinking and Routine Reader Groups in the Orbital Tumor Group

Metric	GPT-5-Thinking at Center 1	Routine Generalists at Center 1	GPT-5-Thinking at Center 2	Routine Subspecialists at Center 2
Benign–malignant accuracy	92.0 (276/300) [88.4, 94.6]	84.7 (254/300) [80.2, 88.3]	92.7 (278/300) [89.2, 95.1]	93.7 (281/300) [90.3, 95.9]
Top-1 accuracy	74.0 (222/300) [68.8, 78.6]	68.7 (206/300) [63.2, 73.7]	76.7 (230/300) [71.6, 81.1]	78.3 (235/300) [73.3, 82.6]
Sensitivity	90.9 (60/66) [81.6, 95.8]	84.8 (56/66) [74.3, 91.6]	92.0 (69/75) [83.6, 96.3]	94.7 (71/75) [87.1, 97.9]
Specificity	92.3 (216/234) [88.2, 95.1]	84.6 (198/234) [79.4, 88.7]	92.9 (209/225) [88.8, 95.6]	93.3 (210/225) [89.3, 95.9]
PPV	76.9 (60/78) [66.4, 84.9]	60.9 (56/92) [50.7, 70.2]	81.2 (69/85) [71.6, 88.1]	82.6 (71/86) [73.2, 89.1]
NPV	97.3 (216/222) [94.2, 98.8]	95.2 (198/208) [91.4, 97.4]	97.2 (209/215) [94.0, 98.7]	98.1 (210/214) [95.3, 99.3]

Note.—This table shows that Generative Pretrained Transformer (GPT)-5-Thinking had higher top-1 accuracy than did routine generalists at center 1 and top-1 accuracy similar to that of routine subspecialists at center 2. Data are percentages; numerator/denominator are in parentheses, and Wilson 95% CIs are in brackets. Phase 1 compared GPT-5-Thinking with one original routine clinical MRI report diagnosis per patient for the same 300 reports at each center. The routine generalist and routine subspecialist comparators were not fixed reader panels; rather, they represented real-world routine clinical report diagnoses generated at center 1 (the Second Hospital of Jilin University) and center 2 (Beijing Tongren Hospital, Capital Medical University), respectively. Sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) refer to malignant classification. Corresponding *P* values for paired benchmarking comparisons were obtained from McNemar tests and are shown in Figure 3A and 3B and Appendix S4.

percentages. For group comparisons, continuous variables were analyzed using independent-samples *t* tests or Mann-Whitney *U* tests. For categorical variables, proportions were compared using χ^2 tests for independent groups and McNemar tests for paired data. In phase 2, paired accuracy outcomes were assessed using mixed-effects logistic regression models, with assistance as a fixed effect and random intercepts for reader and report. Reading time was compared using paired *t* tests. Interreader agreement was evaluated using the Fleiss κ , and differences were assessed using *z* tests (17). All statistical analyses were performed by J.L. and L.X. using SPSS software, version 27.0 (IBM), and R software, version 4.3.0 (R Project for Statistical Computing). A two-sided *P* value less than .05 was considered to indicate a statistically significant difference.

Results

Patient Characteristics

Among the 1717 patients assessed for eligibility, 717 were excluded because of insufficient or discordant histopathologic confirmation (*n* = 482) or absence of an eligible pretreatment index MRI report (*n* = 235) (Fig 1). The final study sample comprised 1000 patients with pathologically confirmed orbital or

head-and-neck tumors who underwent pretreatment MRI (mean age, 51 years ± 17.0; age range, 18–90 years; 520 men), including 600 patients with orbital tumors and 400 with head-and-neck tumors, with 500 patients enrolled at each center (Table 1).

Phase 1: Benchmarking against Routine Clinical Practice

Full phase 1 data are shown in Table 2, Figure 3A–3C, and Table S11. At center 1 (generalist practice), GPT-5-Thinking achieved a benign–malignant accuracy of 92.0% (276 of 300; 95% CI: 88.4, 94.6) compared with 84.7% (254 of 300; 95% CI: 80.2, 88.3) for generalists (*P* = .005). Top-1 accuracy was 74.0% (222 of 300; 95% CI: 68.8, 78.6) versus 68.7% (206 of 300; 95% CI: 63.2, 73.7) for generalists (*P* = .03).

At center 2 (subspecialist practice), GPT-5-Thinking achieved a benign–malignant accuracy of 92.7% (278 of 300; 95% CI: 89.2, 95.1) and top-1 accuracy of 76.7% (230 of 300; 95% CI: 71.6, 81.1) compared with 93.7% (281 of 300; 95% CI: 90.3, 95.9; *P* = .76) and 78.3% (235 of 300; 95% CI: 73.3, 82.6; *P* = .64) for subspecialists, respectively.

Phase 2: Paired Reader-aid Study

Detailed phase 2 results are presented in Table 3, Figures 3D, 3E, 4. The 100-report sample included 40 common typical benign

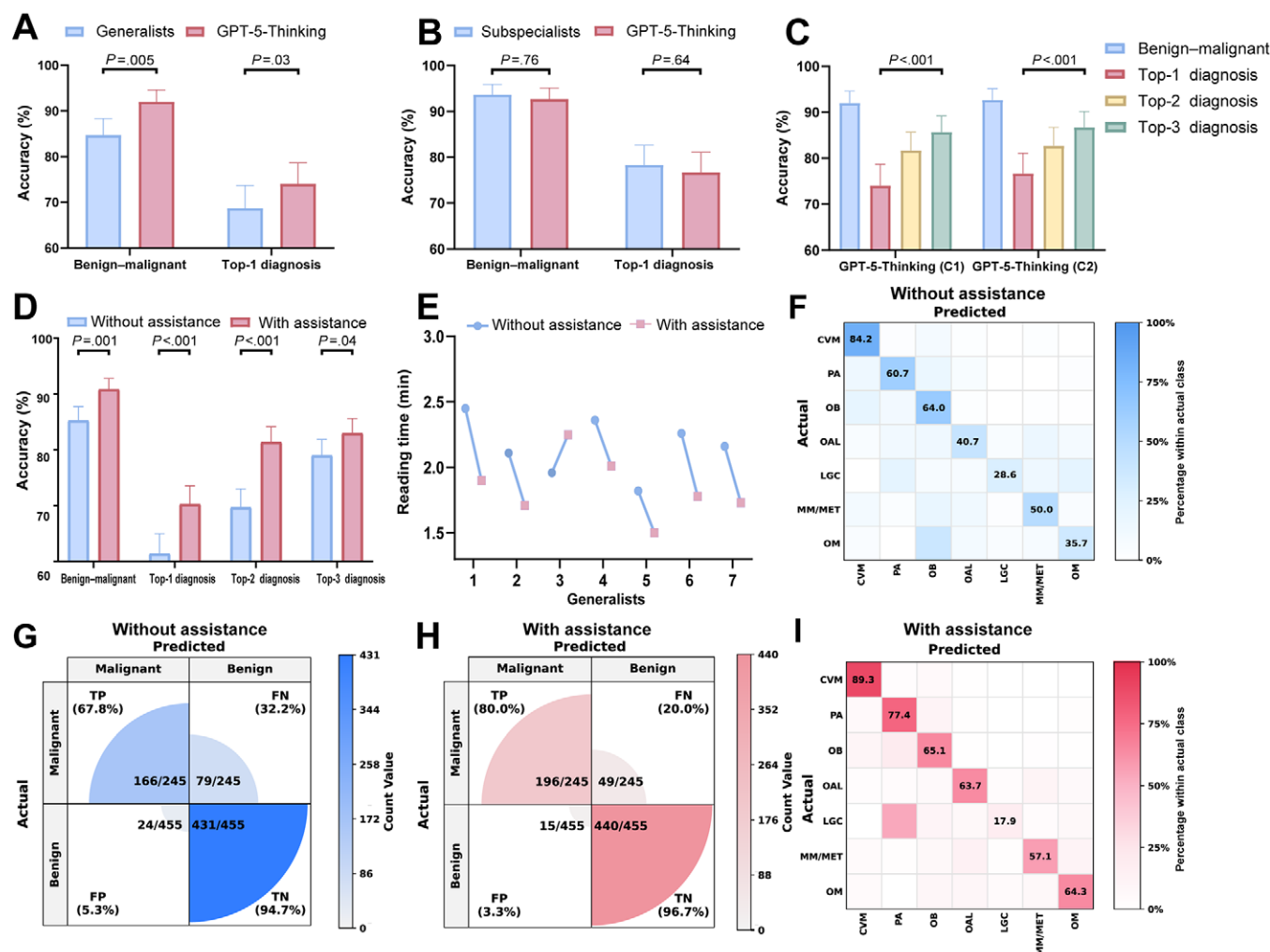


Figure 3: Primary and secondary outcomes of Generative Pretrained Transformer (GPT)-5-Thinking across two study phases. GPT-5-Thinking showed higher diagnostic accuracy than that of generalists at center 1 and accuracy similar to that of subspecialists at center 2; GPT-5-Thinking assistance was associated with higher generalist accuracy and shorter reading time in the reader-aid study. (**A, B**) Phase 1 benchmarking at center 1 and center 2 for benign-malignant classification and top-1 diagnosis. (**C**) GPT-5-Thinking accuracy across benign-malignant classification and top-1, top-2, and top-3 diagnoses at both centers. (**D**) Paired reader-aid accuracy without and with GPT-5-Thinking assistance. (**E**) Paired dot-line plot shows mean reading time for each of the seven generalist readers without and with GPT-5-Thinking assistance. (**F**) Class-wise top-1 heatmap and (**G**) benign-malignant confusion matrix without GPT-5-Thinking assistance. (**H**) Benign-malignant confusion matrix and (**I**) classwise top-1 heatmap with GPT-5-Thinking assistance. In **F** and **I**, rows are normalized within each actual class; in **G** and **H**, percentages are calculated using the actual class as the denominator. Error bars in **A–D** represent Wilson 95% CIs for proportions. P values in **A–C** were obtained from McNemar tests; P values in **D** were obtained from mixed-effects logistic regression models; reading time in **E** was compared using paired t tests. C1 = The Second Hospital of Jilin University, C2 = Beijing Tongren Hospital, Capital Medical University, CVM = cavernous venous malformation, FN = false-negative, FP = false-positive, LGC = lacrimal-gland carcinoma, MM/MET = melanoma/metastasis, OAL = ocular adnexal lymphoma, OB = other benign, OM = other malignant, PA = pleomorphic adenoma, TN = true-negative, TP = true-positive.

tumors, 35 low-frequency malignant tumors, and 25 atypical benign tumors. Across 1400 reader interpretations (700 without GPT-5-Thinking assistance, 700 with assistance), benign-malignant accuracy improved from 85.3% (597 of 700; 95% CI: 82.5, 87.7) without assistance to 90.9% (636 of 700; 95% CI: 88.5, 92.8; $P = .001$) with assistance. Top-1 accuracy increased from 61.4% (430 of 700; 95% CI: 57.8, 65.0) to 70.3% (492 of 700; 95% CI: 66.8, 73.6; $P < .001$). Reading time decreased from 2.16 minutes $\pm$ 0.22 to 1.84 minutes $\pm$ 0.25 (mean difference, -0.32 minutes; $P = .02$). The interreader agreement also rose (Fleiss $\kappa = 0.41$ – 0.71 ; $P < .001$). Report quality scores improved from 2.89 ± 0.20 to 3.46 ± 0.42 ($P < .001$). Regarding secondary metrics, top-2 accuracy rose from 69.7% (488 of 700; 95% CI: 66.2, 73.0) to 81.4% (570 of 700; 95% CI: 78.4, 84.1) (difference, 11.7 percentage points; $P < .001$), and top-3 accuracy increased

from 79.0% (553 of 700; 95% CI: 75.8, 81.9) to 83.0% (581 of 700; 95% CI: 80.0, 85.6) (difference, 4.0 percentage points; $P = .04$).

Extended Diagnostic Evaluation

Category analysis

Without assistance, several malignant categories were frequently misclassified on the classwise heatmap, and the benign-malignant confusion matrix showed correct malignant classifications of 67.8% (166 of 245; 95% CI: 61.7, 73.3) and false-negative classifications of 32.2% (79 of 245; 95% CI: 26.7, 38.3%) (Fig 3F, 3G). With GPT-5-Thinking assistance, correct malignant classifications increased to 80.0% (196 of 245; 95% CI: 74.5, 84.5; $P < .001$), false-negative classifications

Table 3: Phase 2 Diagnostic Metrics and Workflow Outcomes with and without GPT-5-Thinking Assistance in the Orbital Reader Study				
Metric	Without Assistance	With Assistance	Difference	P Value
Benign–malignant accuracy	85.3 (597/700) [82.5, 87.7]	90.9 (636/700) [88.5, 92.8]	5.6	.001
Sensitivity	67.8 (166/245) [61.7, 73.3]	80.0 (196/245) [74.5, 84.5]	12.2	<.001
Specificity	94.7 (431/455) [92.3, 96.4]	96.7 (440/455) [94.6, 98.0]	2.0	.04
PPV	87.4 (166/190) [81.9, 91.4]	92.9 (196/211) [88.6, 95.6]	5.5	.004
NPV	84.5 (431/510) [81.1, 87.4]	90.0 (440/489) [87.0, 92.3]	5.5	.001
Accuracy				
Top 1	61.4 (430/700) [57.8, 65.0]	70.3 (492/700) [66.8, 73.6]	8.9	<.001
Top 2	69.7 (488/700) [66.2, 73.0]	81.4 (570/700) [78.4, 84.1]	11.7	<.001
Top 3	79.0 (553/700) [75.8, 81.9]	83.0 (581/700) [80.0, 85.6]	4.0	.04
Mean reading time (min)	2.16 ± 0.22	1.84 ± 0.25	−0.32	.02
Interreader agreement (Fleiss κ)	0.41	0.71	0.30	<.001
Report quality score	2.89 ± 0.20	3.46 ± 0.42	0.57	<.001

Note.—This table shows that Generative Pretrained Transformer (GPT)-5-Thinking assistance was associated with higher top-1 accuracy, benign–malignant classification accuracy, malignant-lesion sensitivity, interreader agreement, and report quality, and with shorter reading time. Seven generalist readers interpreted the same 100 orbital tumor reports twice, without and with assistance. Percentage values include numerator/denominator in parentheses and Wilson 95% CIs in brackets; reading time and report quality score are means ± SD. Differences are percentage points. Sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) refer to malignant classification. Top-*k* accuracy denotes correct diagnosis within the top *k*-ranked diagnoses. *P* values for benign–malignant accuracy, sensitivity, specificity, PPV, NPV, and top-*k* accuracy were obtained from mixed-effects logistic regression models with assistance as a fixed effect and random intercepts for reader and report. *P* values for reading time and report quality score were obtained from paired *t* tests. The *P* value for interreader agreement was obtained from a *z* test.

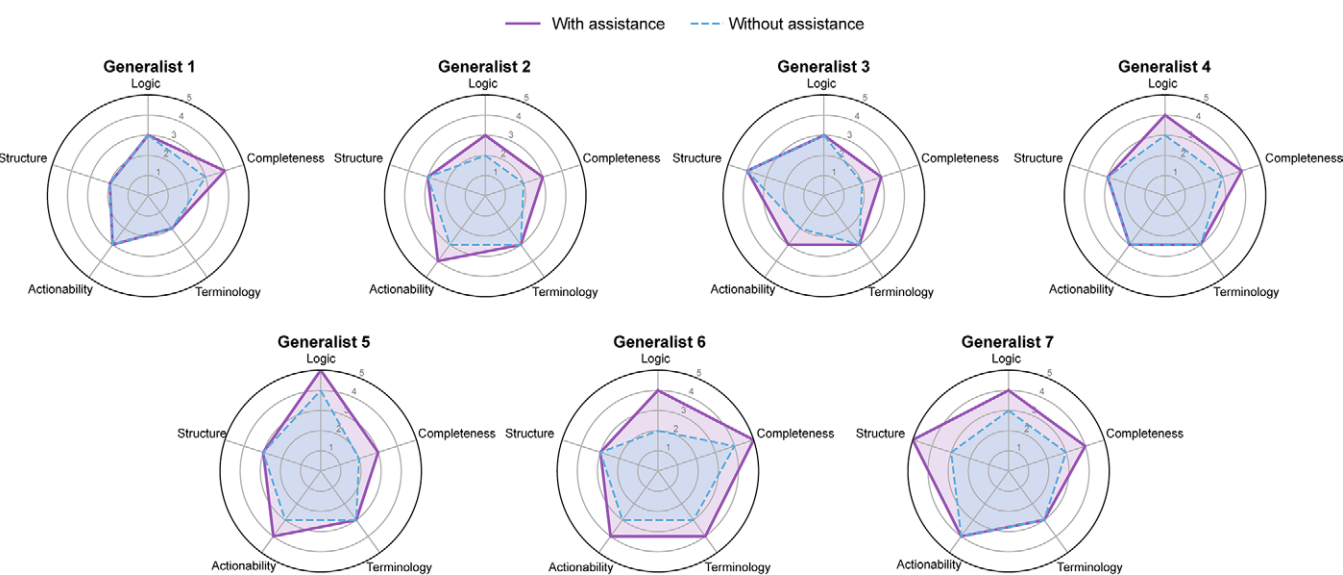


Figure 4: Radar charts of report quality scores across five dimensions for seven generalist radiologists reviewing orbital MRI reports. Generative Pretrained Transformer (GPT)-5-Thinking assistance was associated with higher report quality scores across the five evaluated dimensions for most readers. Blue regions indicate scores without GPT-5-Thinking assistance, and purple regions indicate scores with GPT-5-Thinking assistance. Differences between conditions were analyzed for each individual reader.

decreased to 20.0% (49 of 245; 95% CI: 15.5, 25.5), and the classwise heatmap showed fewer malignant-category misclassifications (Fig 3H, 3I).

Sensitivity analyses

First, in an overlapping accrual-window analysis, 184 eligible patients remained in center 1, and GPT-5-Thinking had higher top-1 accuracy than did contemporaneous generalists (75.5% [139 of 184] vs 69.0% [127 of 184]; *P* = .03). Second, in the heterogeneous tumor subset (*n* = 238 patients, after exclusion of cavernous venous malformation and pleomorphic adenoma), GPT-5-Thinking achieved a top-1 accuracy of 64.7% (154 of

238), compared with 46.2% (110 of 238) for unassisted generalists (*P* < .001) (see Table S12 for details).

Qualitative reasoning analysis

This analysis revealed several key reasoning patterns. First, the model prioritized objective markers over clinical context such as patient age, a pattern that may help counter anchoring bias. Second, it used a location-first hierarchy when signal characteristics conflicted with anatomic constraints. Furthermore, by prompting for subtle signs, the model drew attention to findings that had not been incorporated into some unaided interpretations. Finally, radiologic mimics emerged as a shared

Table 4: Head-and-Neck Extension Diagnostic Metrics for GPT-5-Thinking and Comparator Readers

Analysis or Variable	Patients (n)	GPT-5-Thinking or with Assistance	Comparator or without Assistance	Difference	P Value
Benchmarking: center 1, all reports	200	64.0 (128/200) [57.1, 70.3]	27.0 (54/200) [21.3, 33.5]	37.0	<.001
Larynx and pharynx	64	87.5 (56/64) [77.2, 93.5]	0.0 (0/64) [0.0, 5.7]	87.5	<.001
Major salivary glands	62	58.1 (36/62) [45.7, 69.5]	43.5 (27/62) [31.9, 55.9]	14.5	.11
Sinonasal region and nasopharynx	55	52.7 (29/55) [39.8, 65.3]	36.4 (20/55) [24.9, 49.6]	16.4	.08
Deep neck spaces and skull base	19	36.8 (7/19) [19.1, 59.0]	36.8 (7/19) [19.1, 59.0]	0.0	>.99
Benchmarking: center 2, all reports	200	67.5 (135/200) [60.7, 73.6]	68.0 (136/200) [61.2, 74.1]	−0.5	.92
Reader-aid: top-1 accuracy	100 reports (700 reads)	61.0 (427/700) [57.4, 64.6]	47.0 (329/700) [43.3, 50.7]	14.0	<.001
Reader-aid: benign–malignant accuracy	100 reports (700 reads)	84.0 (588/700) [81.1, 86.5]	76.0 (532/700) [72.7, 79.0]	8.0	.003

Note.—This table shows that Generative Pretrained Transformer (GPT)-5-Thinking had higher top-1 accuracy than routine generalists at center 1 and had top-1 accuracy similar to that of routine subspecialists at center 2; reader accuracy was higher with GPT-5-Thinking assistance than without assistance. Unless otherwise noted, data are percentages; numerator/denominator are in parentheses and Wilson 95% CIs in brackets. In benchmarking, GPT-5-Thinking was compared with one original routine clinical MRI report diagnosis per patient; center 1 was the generalist practice setting and center 2 was the subspecialist practice setting. In the reader-aid analysis, seven generalist readers interpreted 100 head-and-neck tumor reports without and with assistance. *P* values for benchmarking comparisons were obtained from McNemar tests. *P* values for paired reader-aid comparisons were obtained from mixed-effects logistic regression models.

cognitive boundary, where atypical overlapping features led to synchronized failures in both humans and artificial intelligence (see Appendix S5 for details).

Performance ceiling analysis

Subspecialist-adjudicated input optimization yielded the correct diagnosis in 50.0% (39 of 78) of GPT-5-Thinking error reports, suggesting that input insufficiency contributed to model errors. The audit revealed that generalists omitted more descriptors per report (mean, 2.4 ± 1.8) than subspecialists (mean, 1.5 ± 0.8 ; $P < .001$) (see Table S13 for details). To bridge these identified gaps, the Minimum-Descriptor Checklist (Table S14) and the OrbitAssist system (Fig S4, Movie 1) were developed to standardize input quality.

Extension to the Head-and-Neck Group

In the head-and-neck extension group, GPT-5-Thinking showed a pattern of results consistent with that observed in the primary orbital group. In benchmarking against routine clinical practice at center 1, the model had higher accuracy than unassisted generalists (top-1 accuracy, 64.0% [128 of 200; 95% CI: 57.1, 70.3] vs 27.0% [54 of 200; 95% CI: 21.3, 33.5]; $P < .001$), while showing similar top-1 accuracy to subspecialists at center 2 (67.5% [135 of 200; 95% CI: 60.7, 73.6] vs 68.0% [136 of 200; 95% CI: 61.2, 74.1]; $P = .92$) (Table 4). Furthermore, in the paired reader-aid substudy of 100 patients, model assistance was associated with an increase in generalist top-1 accuracy from 47.0% (329 of 700; 95% CI: 43.3, 50.7) to 61.0% (427 of 700; 95% CI: 57.4, 64.6) ($P < .001$) (Table 4; Appendix S7).

Discussion

This dual-center retrospective study evaluated Generative Pretrained Transformer (GPT)-5-Thinking as a report-based diagnostic aid for bridging the diagnostic gap between generalists and subspecialists. In the orbital tumor group, GPT-5-Thinking showed similar top-1 accuracy to subspecialists at the subspecialist

center (76.7% [230 of 300] vs 78.3% [235 of 300]; $P = .64$) and higher top-1 accuracy than routine generalists at the generalist center (74.0% [222 of 300] vs 68.7% [206 of 300]; $P = .03$). GPT-5-Thinking assistance was associated with higher generalist top-1 accuracy in the orbital reader study (61.4% [430 of 700] to 70.3% [492 of 700]; $P < .001$) and in the head-and-neck extension group (47.0% [329 of 700] to 61.0% [427 of 700]; $P < .001$). Interreader agreement was also higher with assistance than without assistance (0.71 vs 0.41; $P < .001$). These findings suggest the potential of reasoning-oriented large language models as subspecialty-informed aids for generalist radiology practice.

Categorical analysis indicated that the observed differences extended beyond benign–malignant binary classification. Binary distinction alone is clinically insufficient because lesions with similar malignant potential may require markedly different management, such as extensive resection versus chemotherapy or radiation therapy (18–20). In practice, radiologists may describe relevant signs but still fail to integrate them into a specific etiologic diagnosis. GPT-5-Thinking appeared to reduce this disconnect by synthesizing anatomic location, enhancement pattern, and disease-specific descriptors into a ranked differential diagnosis.

The sensitivity analyses further supported diagnostic resilience. In the overlapping accrual-window analysis, findings remained stable after exclusion of earlier MRI examinations. In the heterogeneous-subset analysis, GPT-5-Thinking maintained an approximately 20–percentage-point advantage over unassisted generalists after excluding common pathognomonic tumors. This finding is relevant to the long-tail challenge in medical artificial intelligence, where image-based models may generalize poorly to rare or atypical presentations (21,22). Unlike narrowly trained pixel-level models, large language models can use transferable semantic constraints from report text, which may help organize fragmented imaging clues into a coherent diagnostic framework (23,24).

However, this inferential capability remained limited by the quality of the source report, a phenomenon we term the *fidelity paradox* (25–27). Approximately half of model errors were corrected

after re-evaluation with expert-vetted input data, suggesting that incomplete or underspecified observations constrained model reasoning. This finding supports a collaborative interpretation model: When model-derived logic conflicts with the radiologist's initial impression, the discrepancy may prompt targeted reassessment of the original images for subtle but diagnostically important signs. In this setting, the radiologist remains central as the curator of high-fidelity clinical semantics within the human–machine loop (28,29).

Secondary outcomes suggested potential workflow benefit. The higher interreader agreement with assistance was consistent with lower subjective diagnostic variability. Higher report quality and shorter reading times further suggest potential workflow value, although cognitive burden was not directly measured (30). Variation in time savings across readers likely reflected different use patterns, ranging from early diagnostic orientation to late-stage logical verification (31).

The head-and-neck extension group showed that the magnitude of the observed differences varied by anatomic subsite. The largest observed difference occurred in the larynx and pharynx, where diagnostically useful findings were often present in the descriptive report but not translated into a specific final impression. This pattern may reflect conservative reporting in subsites where MRI is commonly used for staging, and definitive characterization is established by endoscopy or biopsy (32,33). In the salivary glands, smaller discrepancies may reflect more characteristic appearances of common tumors (34). In the sinonasal region and nasopharynx, overlapping imaging features likely preserved a modest role for semantic integration (35). In the deep neck spaces and skull base, complex anatomy, greater pathologic heterogeneity, and limited report specificity constrained both human and model inference (36). These subsite patterns reinforce that large language models can reorganize available information but cannot recover decisive evidence absent from the source text.

This study has several limitations. First, the sample was assembled nonconsecutively, by reverse-chronologic ascertainment to fixed per-center targets, which may introduce selection, temporal, and case-mix effects that sensitivity analyses can only partly address. Consequently, the sample may not fully capture the disease spectrum encountered in routine practice; per-center accuracy should be interpreted in the context of this sampling frame, and between-center differences may reflect both sampling and practice-setting differences. Second, because GPT-5-Thinking was used in a reader-directed, nonstandardized manner, the observed gains reflect reader–model interaction and cannot be attributed to the model alone. Third, model output depended on the completeness and specificity of report-level inputs, and incomplete descriptions may constrain inference. These findings are therefore observational and hypothesis-generating, and prospective multicenter studies with standardized reader–model protocols are needed to confirm them.

In summary, in this retrospective report-based evaluation of patients with orbital or head-and-neck tumors, GPT-5-Thinking showed diagnostic accuracy similar to that of subspecialists, and GPT-5-Thinking assistance was associated with higher generalist diagnostic accuracy. Future prospective studies should evaluate whether reasoning-oriented large language models can be integrated with structured reporting and image review workflows to support more consistent subspecialty-informed diagnosis while preserving radiologist oversight.

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